Solutions for Graduate Combinatorics HW

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November 7, 2026

Due to limited time the following are only **sketches** of full solutions, and in particular these solutions alone wouldn't necessarily constitute a solution worth full marks. I also emphasize that there may exist other (possibly simpler) solutions these problems.

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1 HW1

1.1 *Let $\mathcal{F}$ be an r-uniform set system with the property that $F' \cap F'' \not\subseteq F$ for any distinct $F, F', F'' \in \mathcal{F}$. Prove that $|\mathcal{F}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}$ (Hint: let $F'' \in \mathcal{F}$ be arbitrary and consider*

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the set system $\{F \cap F'' : F \in \mathcal{F} \setminus \{F''\}\}$*) [2-].*

The set system $\mathcal{F}''$ as suggested in the hint is an antichain: for distinct $F, F' \in \mathcal{F}$, having $F' \cap F'' \subseteq F \cap F''$ would in particular imply $F' \cap F'' \subseteq F$ which we assumed not to be the case. Moreover, this same argument shows $|\mathcal{F}''| = |\mathcal{F} \setminus \{F''\}|$, i.e. that $F \cap F'' \neq F' \cap F''$ for distinct F, F'. Since $\mathcal{F}''$ has ground set F'' of size r, it follows from Sperner's Theorem that $|\mathcal{F}''| \leq \binom{r}{\lfloor r/2 \rfloor}$, and hence

$$|\mathcal{F}| = 1 + |\mathcal{F} \setminus \{F''\}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}.$$

1.5 *Prove* **??** *(Hint: try proving the weaker lower bound of* n^{-2^d} *to get a feel for the proof idea) [2].*

The case $d = 1$ is Dilworth's Theorem. Inductively assume we have proven the result up to some value $d \geq 2$. Let G Be a graph and $P_1, \ldots, P_d$ posets on $V(G)$ such that $x \sim y$ iff $x <_i y$ or $y <_i x$ for some i. If P_d contains a chain on at least $n^{1/(d+1)}$ elements then this corresponds to a clique of size at least $n^{1/(d+1)}$ in G. By the Corollary of Dilwrorth's Theorem that $n \leq h(P)w(P)$ for all P, we can assume P_d contains an antichain on at least $n^{d/(d+1)}$ vertices of G. Let $G' \subseteq G$ be the induced subgraph of G on this antichain. Since G' is induced on an antichain of P_d, it follows that $P_1, \ldots, P_{d-1}$ are such that two vertices of G' are adjacent if and only if they are comparable by some P_i with $1 \leq i \leq d-1$, so inductively G' has either a clique or independent set of size at least $(n^{d/(d+1)})^{1/d} = n^{1/(d+1)}$, proving the result.

1.7a *Prove Erdős-Szekeres using Dilworth's Theorem [2].*

Construct a poset P on $[n]$ where $i <_P j$ if and only if $a_i < a_j$ and $i < j$ (with these latter inequalities being with respect to the usual ordering on $\mathbb{R}$). By Dilworth there exists a chain or an antichain of size at least $\sqrt{n}$ of P, and it is straightforward to check that these correspond to increasing or decreasing subsequences of π of the same size.

1.7b *Prove this without using Dilworth's Theorem (Hint: for each* j *define* i_j *to be the length of the longest increasing subsequence which ends at* a_j *and* d_j *to be the length of the longest decreasing subsequence which ends at* a_j*. Can we have* $(i_j, d_j) = (i_{j'}, d_{j'})$ *for* $j < j'$*?) [2+].*

Following the hint, we can not have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$: if $a_j < a_{j'}$ then we could add $a_{j'}$ to the end of the increasing subsequence of length i_j ending at a_j to find $i_{j'} > i_j$; an analogous argument works if $a_j > a_{j'}$. It follows that the n pairs (i_j, d_j) are distinct, which means we can not have $i_j, d_j < \sqrt{n}$ for all n and hence some i_j or d_j is at least this large, corresponding to a large increasing or decreasing subsequence.

1.8 *Prove Mirsky's Theorem: if* $P = (X, \leq)$ *is a finite poset of height* $h(P)$*, then there exists a partition of* X *into* $h(P)$ *antichains (Hint: prove that the set of maximal elements* M *of* P *form an antichain) [2].*

For the hint, if two maximal elements x, y were comparable, say with $x < y$, then x would not be maximal. We conclude that M is indeed an antichain.

We now prove the main result by induction on $h(P)$, the case $h(P) = 0$ being trivial since this can only occur if $P = \emptyset$. We claim that $P - M$ has height at most $h(P) - 1$; indeed, if $P - M$ had a chain of length $h(P)$ then whatever element is at the top of the chain is not maximal and hence could be extended to a chain of length $h(P) + 1$ in P, a contradiction. Inductively then we can decompose $P - M$ into at most $h(P) - 1$ antichains, which together with M gives the desired partition.

2 HW2

1.2b *Prove for any integer $t \geq 1$ that*

$$\sup_{r \in \mathbb{R}} \Pr[\sum_i a_i X_i \in [r, r+2t)] \leq \max_{T \subseteq \binom{\{0,1,\ldots,n\}}{t}} \sum_{r \in t} \binom{n}{r} 2^{-n},$$

where to be clear this upper bound is equal to 2^{-n} times the sum of the t largest binomial coefficients [2].

As in the Littlewood-Offord proof, we may assume $a_i > 0$ for all i. Fix an arbitrary r and let $\mathcal{F}$ be the set of $F \subseteq [n]$ with the property that $\sum_{i \in F} a_i - \sum_{i \notin F} a_i \in [r, r+t+1)$. By Erdős's Theorem it suffices to prove that $\mathcal{F}$ does not have a chain of size larger than t. Say for contradiction that it contained some $F_1 \subsetneq \cdots \subsetneq F_{t+1}$. Define $\sigma_j = \sum_{i \in F_j} a_i - \sum_{i \notin F_j} a_i$. It is straightforward to prove by induction that $\sigma_j \geq 2(j-1) + \sigma_1$ since each a_i is at least 1. In particular since $\sigma_1 \geq r$ by hypothesis of $F_1 \in \mathcal{F}$ we find $\sigma_{t+1} \geq 2t + r$, a contradiction to $F_{t+1} \in \mathcal{F}$ implying $\sigma_{t+1} < 2t + r$.

2.1 *We consider an abstract version of the Katona circle argument: for a permutation $\pi : [n] \to [n]$, we define for $F = \{x_1, \ldots, x_r\} \subseteq [n]$ the set $\pi(F) = \{\pi(x_1), \ldots, \pi(x_r)\}$ and for $\mathcal{F} \subseteq 2^{[n]}$ we define $\pi(\mathcal{F}) = \{\pi(F) : F \in \mathcal{F}\}$. Prove that for any two families $\mathcal{F}, \mathcal{A} \subseteq 2^{[n]}$ that*

$$\sum_{F \in \mathcal{F}} \frac{\left|\mathcal{A} \cap \binom{[n]}{|F|}\right|}{\binom{n}{|F|}} \leq \max_\pi |\pi(\mathcal{F}) \cap \mathcal{A}|,$$

where the maximum ranges over all permutations $\pi : [n] \to [n]$ [2-].

We count all pairs (π, F) with $\pi(F) \in \mathcal{A}$ in two ways. Each of the $n!$ bijections π is trivially in at most $\max_\pi |\pi(\mathcal{F}) \cap \mathcal{A}|$ pairs. On the other hand, for each F we can identify (π, F) by specifying the image $\pi(F) \in \mathcal{A} \cap \binom{[n]}{|F|}$ and given this we have $|F|!(n-|F|)! = \frac{n!}{\binom{n}{|F|}}$ ways to specify π. Setting these two counts equal to each other and dividing both sides by $n!$ gives the stated bound.

2.2 *Give an alternative proof of the LYM inequality by using that if $\mathcal{I}$ is the set of initial intervals $\{1, 2, \ldots, i\}$ for $1 \leq i \leq n$ then every antichain of $\mathcal{I}$ has size at most 1 [2-].*

This follows from the previous exercise applied with $\mathcal{A} = \mathcal{I}$ after observing that if $\mathcal{F}$ is an antichain, then at most one $F \in \mathcal{F}$ can have $\pi(F)$ an initial interval (since a second F' with, say, $\pi(F')$ an initial interval containing the initial interval $\pi(F)$ would imply $F \subseteq F'$).

2.5 *We say that a family $\mathcal{F} \subseteq 2^{[n]}$ is union if $F \cup F' \neq [n]$ for all $F, F' \in \mathcal{F}$. Prove that $\mathcal{F}$ is a union family if and only if*

$$\overline{\mathcal{F}} := \{[n] \setminus F : F \in \mathcal{F}\}$$

is intersecting. In particular, any problem about intersecting families can be turned into a problem about union families, which in practice can sometimes be easier to solve [1+].

Any $x \in F \cap F'$ satisfies $x \notin ([n] \setminus F) \cup ([n] \setminus F') = [n] \setminus (F \cap F')$, giving the result.

3 HW3

2.6a *Prove that every maximal intersecting family $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| = 2^{n-1}$. For this we emphasize that by definition, no intersecting family $\mathcal{F}$ can have $\emptyset \in \mathcal{F}$ since then $F \cap F' = \emptyset$ for $F = F' = \emptyset$ elements in $\mathcal{F}$ [2].*

Let $\mathcal{F}$ be maximal intersecting. We prove the stronger fact that for every $F \subseteq [n]$ exactly 1 of F or $[n] \setminus F$ is in $\mathcal{F}$. Certainly at most 1 of these sets can be in $\mathcal{F}$ if it is intersecting, so now assume for contradiction that there is some F such that neither is in $\mathcal{F}$. If $\{F\} \cup \mathcal{F}$ is intersecting then this contradicts $\mathcal{F}$ being maximal, so there must be some $F' \in \mathcal{F}$ disjoint from F. This means $F' \subseteq [n] \setminus F$, and this in turn means that $\{[n] \setminus F\} \cup \mathcal{F}$ is intersecting since any F'' disjoint from this would be disjoint with F'.

2.6b *Prove that there exist maximal intersecting families $\mathcal{F} \subseteq \binom{[n]}{r}$ with $|\mathcal{F}| \leq 4^r$ for all n, r [2].*

We trivially have $|\mathcal{F}| \leq 2^n$ for any family $\mathcal{F}$ so it suffices to prove the result when $n \geq 2r$. There are many possible solutions, the simplest is to take $\mathcal{F} = \binom{[2r-1]}{r}$ which is intersecting but any $F \not\subseteq [2r-1]$ will necessarily intersect $[2r-1]$ in at most $r-1$ vertices and hence be disjoint from some edge in $\mathcal{F}$, so $\mathcal{F}$ is maximal with size at most $2^{2r-1} \leq 4^r$.

2.8 *Recall that for a family $\mathcal{F} \subseteq 2^{[n]}$ we define its Lubell measure*

$$\lambda(\mathcal{F}) = \sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}}.$$

Prove that if $\mathcal{F}$ is intersecting then $\lambda(\mathcal{F}) \leq \frac{n+1}{2}$ [2].

Proof without circle method: The key idea is that for each F, at most one of F and $[n] \setminus F$ is in $\mathcal{F}$. The slick way of doing this is:

$$2\lambda(\mathcal{F}) = \sum_{F \in \mathcal{F}} 2\binom{n}{|F|}^{-1} = \sum_{F \in \mathcal{F}} \binom{n}{|F|}^{-1} + \binom{n}{|[n] \setminus F|}^{-1} \leq \sum_{F \in 2^{[n]}} \binom{n}{|F|}^{-1} = n+1,$$

with the last equality using that each of the $n+1$ layers has Lubell weight 1.

Proof with circle method: Take $\mathcal{A}_n^*$ to be the family of all cyclic intervals except for $[n]$. Any $\mathcal{F} \subseteq \mathcal{A}_n^*$ which is intersecting has size at most $|\mathcal{A}_n^*| = \binom{n}{2}$ again simply because an

intersecting family can not contain both a set and its complement. Applying the abstract Katona circle argument from Exercise 2.1 then yields that any intersecting $\mathcal{F} \subseteq 2^{[n]} \setminus [n]$ has $\lambda(\mathcal{F}) = \sum_F \binom{n}{|F|}^{-1} \leq \frac{n-1}{2}$, and this in turn implies that any intersecting $\mathcal{F} \subseteq 2^{[n]}$ has $\lambda(\mathcal{F}) \leq 1 + \frac{n-1}{2}$ since $\lambda([n]) = 1$.

We emphasize that a direct argument with $\mathcal{A}_n$ instead of $\mathcal{A}_n^*$ does not quite work out because $|\mathcal{A}_n \cap \binom{[n]}{n}$ does not equal n like in all the other terms but rather 1.

4 HW4

3.8 *Prove that for every family $\mathcal{F}_1 \subseteq 2^{[n]}$ there is some finite sequence of integers $i_1, \ldots, i_t$ such that if we define $\mathcal{F}_{k+1} = s_{-i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is a downset [2-].*

Similar to our proof of **??** we take $\mathcal{F}_{k+1} = s_{-i_k}(\mathcal{F}_k)$ whenever there exists i_k with $\mathcal{F}_k \neq s_{-i_k}(\mathcal{F}_k)$. One can easily check then that the quantity

$$\sum_{e \in \mathcal{F}_k} |e|$$

strictly decreases in k, so at some point this process must stop at some $\mathcal{F}_{t+1}$ which has $\mathcal{F}_{t+1} = s_{-i}(\mathcal{F}_{t+1})$ for all i, and it is straightforward to check that this implies $\mathcal{F}_{t+1}$ is a downset.

3.10 *We say a family $\mathcal{F}$ has diameter d if $\max_{F,F' \in \mathcal{F}} |F \Delta F'| = d$ where here $F \Delta F'$ denotes the symmetric difference, and we let $\operatorname{diam}(\mathcal{F})$ denote the diameter of $\mathcal{F}$. Prove that $\operatorname{diam}(s_{-i}(\mathcal{F})) \leq \operatorname{diam}(\mathcal{F})$ for all families $\mathcal{F}$ and integers i [2-].*

Let $F, F' \in \mathcal{F}$ be such that $\operatorname{diam}(s_{-i}(\mathcal{F})) = |s_{-i}(F) \Delta s_{-i}(F')|$. If i is not in this symmetric difference then $|s_{-i}(F) \Delta s_{-i}(F')| \leq |F \Delta F'|$ since the only element which possibly changed in these sets is i, and from this it follows that the diameter of $\mathcal{F}$ is at least that of $s_{-i}(\mathcal{F})$. As such we may assume that, say, $i \in s_{-i}(F)$ and $i \notin s_{-i}(F')$. The fact that $i \in s_{-i}(F)$ implies that $F - i \in \mathcal{F}$. This implies that $|s_{-i}(F) \Delta s_{-i}(F')|$ is equal to either $|F \Delta F'|$ or $|(F - i) \Delta F'|$ depending on if $i \in F'$ or not. In either case this shows $\operatorname{diam}(s_{-i}(\mathcal{F})) = |s_{-i}(F) \Delta s_{-i}(F')| \leq \max\{|F \cap F'|, |(F - i) \cap F'|\} \leq \operatorname{diam}(\mathcal{F})$, proving the result.

3.11a *Prove Katona's Union Theorem using Kleitman's Diameter Theorem [1+].*

If $\mathcal{F}$ is such that $|F \cup F'| \leq d$ then certainly $|F \Delta F'| \leq d$, so the bound for Katona follows from the bound of Kleitman.

3.11b *Prove Kleitman's Diameter Theorem using Katona's Union Theorem [2].*

By 3.8 and 3.10 we can assume $\mathcal{F}$ is a downset. We claim then that $\mathcal{F}$ is $(n-d)$-union. Indeed, if we had $|F \cup F'| > d$ then because $\mathcal{F}$ is a downset we have $F'' := F' \setminus F \in \mathcal{F}$, and this satisfies $|F \Delta F''| = |F \cup F'| > d$, a contradiction. We conclude that $\mathcal{F}$ is $(n-d)$-union, so the bound follows from Katona.

5 HW5

3.6 *Given a graph G, let $\kappa(G)$ denote the total number of cliques of G, i.e. this is the number of K_1's of G plus the number of K_2's of G, and so on. For integers m, n with $m = \binom{k}{2}$ for some integer $k \leq n$, determine the largest possible value of $\kappa(G)$ amongst n-vertex graphs with m edges [2+].*

We claim that the answer is $\sum_{i=1}^{k} \binom{k}{i}$. The lower bound comes from a clique on k vertices. For the upper bound, we claim that any graph G with $\binom{k}{2}$ edges has at most $\binom{k}{i}$ cliques of size i. Indeed, let $\mathcal{F}$ denote the set system on $V(G)$ whose hyperedges are cliques of size i. Then $\partial_2 \mathcal{F} \subseteq E(G)$ and hence $|\partial_2 \mathcal{F}| \leq \binom{k}{2}$, which by the contrapositive of the Lovász version of Kruskal-Katona implies $|\mathcal{F}| \leq \binom{k}{i}$, proving the claim and hence the result.

4.2 *Prove for all $r, p \geq 1$ that there exists an r-uniform set system $\mathcal{F}$ with $|\mathcal{F}| = (p-1)^r$ which does not contain a sunflower with p petals [2-].*

The result is trivial for $p = 1$ so assume $p > 1$ from now on. Let $X_1, \ldots, X_r$ be disjoint sets on $p-1$ elements and let $\mathcal{F}$ be the set system consisting of all sets which use exactly 1 vertex from each X_i. We claim this does not have a sunflower with p petals. Assume otherwise with $S_1, \ldots, S_p$ the edges and K the kernel. We must have $|K| < r$ (else $S_i = S_j$ for all i, j, giving a sunflower with $1 < p$ edges), and in particular K is disjoint from some X_i. By construction each edge S_j must use exactly one vertex of X_i and these two vertices can not equal each other since otherwise the sunflower condition would force this vertex in K, a contradiction to K being disjoint from X_i. However, it is impossible for each S_j to use distinct vertices from X_i since $|X_i| < p$, and this gives the desired contradiction.

4.4a *Prove Deza's Theorem when $\ell = 1$ (Hint: prove that if $\mathcal{F}$ contains a sufficiently large sunflower, then $\mathcal{F}$ must be a sunflower) [2].*

We claim that if $\mathcal{F}$ contains a sunflower with at least $r+1$ petals then $\mathcal{F}$ is a sunflower. Indeed, say $\mathcal{F}$ had a largest sunflower with edges $S_1, \ldots, S_p$ for $p \geq r+1$, and by hypothesis this sunflower must have a kernel of size 1, say with $K = \{v\}$. Assume there was some edge $F \in \mathcal{F}$ not part of this sunflower. If $v \in F$ then F by hypothesis must be disjoint from $S_i \setminus \{v\}$ for all i, meaning F can be added to the sunflower to make a larger one. Thus $v \notin F$, which again by hypothesis means F must intersect each of the disjoint sets $S_i \setminus \{v\}$ which is impossible for $p \geq r+1$.

With this claim in mind, consider any $F = \{v_1, \ldots, v_r\} \in \mathcal{F}$. By a similar argument to the above we observe that the set of edges of $\mathcal{F}$ containing each v_i form a sunflower, so by the claim we may assume that the number of edges containing each v_i is at most r (i.e. there are at most $r-1$ such edges other than F itself). Since every edge contains at least one v_i by hypothesis on $\mathcal{F}$, we find that the total number of edges of $\mathcal{F}$ is at most $1 + r(r-1) = r^2 - r + 1$, contradicting our size condition on $\mathcal{F}$.

6 HW6

5.1 *For all integers $r, \nu \geq 1$ construct an r-uniform $\mathcal{F}$ with $\nu(\mathcal{F}) = \nu$ and $\tau(\mathcal{F}) = r\nu$ [2-].*

Take $K^{(r)}_{r\nu+r-1}$ which has matching number ν and transversal number $r\nu$.

5.2 *Prove that if $\mathcal{F}$ is an intersecting family then $\tau(\mathcal{F}) \leq \min_{F\in\mathcal{F}} |F|$ [1+].*

We observe that any edge $F \in \mathcal{F}$ is a transversal since by assumption every edge of $\mathcal{F}$ intersects F, giving the bound.

5.3 *(Gyárfás) Prove that if $\mathcal{F}$ is a set system where every edge has size at most r, then for every set of vertices S the number of transversals T of $\mathcal{F}$ with $S \subseteq T$ and $|T| = \tau(\mathcal{F})$ is at most $r^{\tau(\mathcal{F})-|S|}$ [2].*

We prove the result by backwards induction on S. The base case of $|S| = \tau(\mathcal{F})$ is trivial. Assume that $|S| < \tau(\mathcal{F})$ and that we have proven the result for all sets larger than S. Because $|S| < \tau(\mathcal{F})$ there is some $F \in \mathcal{F}$ which does not intersect F. Observe that every transversal which contains S must contain $S \cup \{v\}$ for some $v \in F$. Inductively the number of T containing a given $S\cup\{v\}$ is at most $r^{\tau(\mathcal{F})-|S|-1}$, and summing this over each of the $|F| \leq r$ choices for v gives at most $r^{\tau(\mathcal{F})-|S|}$ choices for T, completing the inductive step.

5.4 *(Erdős-Lovász) Prove that if $\mathcal{F}$ is an r-uniform intersecting family with $\tau(\mathcal{F}) = r$ then $|\mathcal{F}| \leq r^r$ [2].*

By the previous problem applied with $S = \emptyset$ there are at most r^r transversals of $\mathcal{F}$ of size r. Because $\mathcal{F}$ is intersecting and r-uniform, every edge is a transversal of size r, so in total this means $|\mathcal{F}| \leq r^r$.

7 HW7

5.5a *Prove for all i and sets of vertices X that $|\mathrm{Tr}_{s_{-i}(\mathcal{F})}(X)| \leq |\mathrm{Tr}_{\mathcal{F}}(X)|$ [2].*

If $i \notin X$ then $F \cap X = s_{-i}(F) \cap X$ for every $F \in \mathcal{F}$ and the traces equal each other, so from now on we may assume $i \in X$. We will show that for every $T \in \mathrm{Tr}_{s_{-i}(\mathcal{F})}(X) \setminus \mathrm{Tr}_{\mathcal{F}}(X)$ that $i \notin T$ and $T \cup \{i\} \in \mathrm{Tr}_{\mathcal{F}}(X) \setminus \mathrm{Tr}_{s_{-i}(\mathcal{F})}(X)$, and hence there exists an injection from $\mathrm{Tr}_{s_{-i}(\mathcal{F})}(X)$ to $\mathrm{Tr}_{\mathcal{F}}(X)$ by mapping any T in both sets to itself and otherwise mapping T to $T \cup \{i\}$.

To this end, for each $T \in \mathrm{Tr}_{s_{-i}(\mathcal{F})}(X) \setminus \mathrm{Tr}_{\mathcal{F}}(X)$ let $F' \in s_{-i}(\mathcal{F})$ be such that $F' \cap X = T$. Observe that we must have $F' \notin \mathcal{F}$ as otherwise we would have $T \in \mathrm{Tr}_{\mathcal{F}}(X)$ as well, so this implies F' is a squashed edge, meaning $F' \cup \{i\} \in \mathcal{F}$ and $i \notin F'$. In particular, $T = F' \cap X$ can not contain i and also $(F' \cup \{i\}) \cap X = T \cup \{i\} \in \mathrm{Tr}_{\mathcal{F}}(X)$. Thus we are done unless also $T \cup \{i\} \in \mathrm{Tr}_{s_{-i}(\mathcal{F})}(X)$. Say this were true with $F'' \in s_{-i}(\mathcal{F})$ such that $F'' \cap X = T \cup \{i\}$. This means $i \in F''$ which by definition of $s_{-i}(\mathcal{F})$ implies $F'' \setminus \{i\} \in \mathcal{F}$, and hence that $(F'' \setminus \{i\}) \cap X = T \in \mathrm{Tr}_{\mathcal{F}}(X)$, a contradiction.

5.5b *Use this to prove the Sauer-Shelah lemma (Hint: use the exercises from **??** to compress $\mathcal{F}$ to a downset) [2-].*

According to the hint, we can perform a sequence of squashing operations to $\mathcal{F}$ to turn it into a downset $\mathcal{F}'$. We claim that $\mathcal{F}'$ has VC dimension at most that of $\mathcal{F}$. Indeed, if a set X is shattered in $\mathcal{F}'$ then the previous exercise implies it was also shattered in X.

Observe that in a downset $\mathcal{F}'$ that a set is shattered if and only if it is an edge of $\mathcal{F}'$, so if $\mathcal{F}$ (and hence $\mathcal{F}'$) has VC dimension at most d, this implies that $\mathcal{F}'$ only contains sets of size at most d, which means $|\mathcal{F}| = |\mathcal{F}'| \leq \sum_{i=0}^{d} \binom{|V|}{i}$, proving the result.

5.6 *Prove that if $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| > 1 + n + n^2/4$, then there exists some $S \subseteq [n]$ of size 3 with $|\mathrm{Tr}(S)| \geq 7$ (Hint: reduce to $\mathcal{F}$ being a downset as in the previous problem and use Mantel's Theorem from graph theory) [2-].*

Again we can assume by applying a sequence of squashing operations (which can only make it harder to find a set with $|\mathrm{Tr}(S)| \geq 7$) that $\mathcal{F}$ is a downset. If $F \in \mathcal{F}$ has $|F| \geq 3$ then any $S \subseteq F$ is shattered, so we can assume every element of $\mathcal{F}$ has size at most 2. If $|\mathcal{F}| > 1 + n + n^2/4$ then this implies $\mathcal{F}$ contains more than $n^2/4$ sets of size 2. By Mantel's Theorem this implies $\mathcal{F}$ contains three sets of size 2 which form a graph triangle on some set $S = \{x, y, z\}$, and again $\mathcal{F}$ being a downset implies $|\mathrm{Tr}(S)| \geq 7$, proving the result.

8 HW8

5.7a *Prove that if $\mathcal{F}'$ is a set system obtained by adding some number of edges of size at most k to $\mathcal{F}$, then $\mathcal{F}'$ has VC dimension at most $d + k + 1$ [2].*

Assume for contradiction that $\mathcal{F}'$ shattered some set X of size $d + k + 2$. Let $K \subseteq X$ be an arbitrary set of size $k + 1$. We claim that the set $X - K$ is shattered in $\mathcal{F}$. Indeed, for every subset $S \subseteq X - K$ there exists $F \in \mathcal{F}'$ with $F \cap X = S \cup K$ since X is shattered in $\mathcal{F}'$, and this means $F \cap (X - K) = S$. Moreover, $|F| \geq |K| > k$, which means $F \in \mathcal{F}$ since $\mathcal{F}' \setminus \mathcal{F}$ only has sets of size at most k. By considering these F for each possible S shows that indeed $X - K$ is shattered in $\mathcal{F}$. But $|X - K| = d + 1$, a contradiction to the VC-dimension of $\mathcal{F}$.

6.2b *(Triangular Criterion) Prove that if for all j there exists some $s_j \in S$ with $p_i(s_j) = 0$ for $i < j$ and $p_j(s_j) \neq 0$ then $p_1, \ldots, p_m$ are linearly independent over $\mathbb{F}$ [1+].*

Let $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ be such that $\sum \lambda_i p_i = 0$. We claim that $\lambda_j = 0$ for all j. Assume this were false, which means there exists some largest j such that $\lambda_j \neq 0$, and in particular we only have to consider the sum from 1 to j. Plugging in s_j into both sides of the equality gives $0 = \sum_{i=1}^{j} \lambda_i p_i(s_j) = \lambda_j p_j(s_j) \neq \emptyset$, a contradiction.

6.3b *(Eventown) Determine the largest size of a family $\mathcal{F} \subseteq 2^{[n]}$ satisfying $|F| \equiv 0 \mod 2$ for $F \in \mathcal{F}$ and $|F \cap F'| \equiv 0 \mod 2$ for distinct $F, F' \in \mathcal{F}$ (Hint: let V denote the span of the characteristic vectors of $\mathcal{F}$ over $\mathbb{F}_2^n$ and show that $V \subseteq V^{\perp}$; what does that say about the dimension of V?) [2].*

We claim that the answer is $2^{\lfloor n/2 \rfloor}$. For the construction we can simply consider the collection $\mathcal{F}$ of all sets F with the property that $2i-1 \in \mathcal{F}$ if and only if $2i \in F$ for all $1 \leq i \leq \lfloor n/2 \rfloor$ which is easily seen to satisfy the conditions.

For the upper bound, we let V be as in the hint. Note that $\langle \chi_F, \chi_{F'} \rangle$ equals $|F \cap F'| \mod 2$ which is 0 for all $F, F' \in \mathcal{F}$, so every vector in V is orthogonal to every vector of V and hence $V \subseteq V^\perp$. By rank nullity we have $n = \dim(\mathbb{F}_2^n) = \dim(V) + \dim(V^\perp) \geq 2\dim(V)$ so $\dim(V) \leq \lfloor n/2 \rfloor$. Note that $|\mathcal{F}| = |V| = 2^{\dim(V)} \leq 2^{\lfloor n/2 \rfloor}$, proving the result.

9 HW9

6.4a *Prove that if each $F \in \mathcal{F}$ has $|F| \geq \ell+1$ then $|\mathcal{F}| \leq n$ (Hint: show that the characteristic vectors χ_F are linearly independent by showing $\langle \sum_F \lambda_F \chi_F, \sum_F \lambda_F \chi_F \rangle > 0$ whenever the λ_F are not all 0) [2+].*

Proceeding as in the hint we observe that this inner product equals

$$\sum_F \lambda_F^2 |F| + \sum_{F \neq F'} \lambda_F \lambda_{F'} |F \cap F'| = \sum_F \lambda_F^2 |F| + \sum_{F \neq F'} \lambda_F \lambda_{F'} \ell$$

$$= \sum_F \lambda_F^2 (|F| - \ell) + \sum_{F,F'} \lambda_F \lambda_{F'} \ell = \sum_F \lambda_F^2 (|F| - \ell) + \ell (\sum_F \lambda_F)^2 \geq \sum_F \lambda_F^2,$$

where here this last step used $|F| \geq \ell + 1$ (and that the coefficient λ_F^2 is non-negative), and from this the claim follows. Alternatively one could more directly use $|F| \geq \ell+1$ via noting

$$\sum_F \lambda_F^2 |F| + \sum_{F \neq F'} \lambda_F \lambda_{F'} |F \cap F'| \geq \sum_F \lambda_F^2 (\ell+1) + \sum_{F \neq F'} \lambda_F \lambda_{F'} \ell$$

$$= \sum_F \lambda_F^2 + \ell (\sum_F \lambda_F)^2 \geq \sum_F \lambda_F^2.$$

From the claim it follows that the characteristic vectors are independent in $\mathbb{R}^n$, and hence there are at most n of them, proving $|\mathcal{F}| \leq n$.

6.4b *(Non-uniform Fischer inequality) conclude that all $\{\ell\}$-intersecting families $\mathcal{F} \subseteq 2^{[n]}$ with $\ell, n \geq 1$ have $|\mathcal{F}| \leq n$ [1].*

If every set has size at least $\ell+1$ then it follows from the previous exercise. Otherwise say $F \in \mathcal{F}$ has at most ℓ elements. Because every other $F' \in \mathcal{F} \setminus \{F\}$ intersects F in exactly ℓ vertices, either $\mathcal{F} = \{F\}$ and we are done or $|F| = \ell$ and every element of $\mathcal{F}$ contains F. Moreover, again the fact that any two edges intersect in ℓ vertices mean that the sets $F' \setminus F$ must be disjoint for every $F' \in \mathcal{F}$. From this it follows that $|\mathcal{F}| \leq 1 + n - \ell \leq n$.

8.2 *For two sets $A, B \subseteq \mathbb{Z}_p$ we define $A \oplus B = \{a+b : a \in A,\ b \in B,\ a \neq b\}$. Prove that if p is prime, then $|A \oplus B| \geq \min\{|A| + |B| - 3, p\}$ (Hint: construct a polynomial $f(x,y)$ which always vanishes when $x = y$) [2].*

Consider the polynomial $f(x,y) = (x-y)\prod_{c\in A\oplus B}(x+y-c)$. By construction we have $f(A,B) = 0$ and $\deg(f) = 1 + |A \oplus B|$, and essentially mimicking the proof for Cauchy-Davenport we derive the result This ultimately works but the proof details are more complicated than I intended, need to adjust.

10 HW10

9.1a *Prove that for any (possibly infinite) family $\mathcal{C}$ of convex subsets of $\mathbb{R}^d$ that $\bigcap_{C\in\mathcal{C}} C$ is convex, and conclude that for any $S \subseteq \mathbb{R}^n$ that there exists a unique smallest convex set containing S [1+].*

If $x, y \in \bigcap C$ then the line segment from x to y is in C for each $C \in \mathcal{C}$ since C is convex and $x, y \in C$, and as such $\bigcap C$ will also contain this line segment since every element of the intersection contains this line segment. From this, for any set S if we let $\mathcal{C}$ denote the set of convex sets containing S then $C' := \bigcap_{C\in\mathcal{C}} C$ is a convex set which contains S since $S \subseteq C$ for all C in the intersection. Moreover, any convex set C containing S must contain C' by definition of how C' was constructed, which implies that C' is the unique smallest convex set containing S.

9.1b *Prove that if $C \subseteq \mathbb{R}^d$ is convex then it contains all of its convex combinations [2-].*

We want to prove $\sum_{i=1}^n \lambda_i v_i \in C$ for all possible convex combinations, and we do this by induction on n, the $n = 1$ case being trivial. Assume then that we have proven the result up to some value $n \geq 2$. If some $\lambda_i = 0$ then the point can be written as a convex combination of fewer points, so we can assume $\lambda_i > 0$ for all i. We can write any convex combination with $n \geq 2$ as $\lambda_n v_n + (1-\lambda_n)(\sum_{i=1}^{n-1} \frac{\lambda_i}{1-\lambda_n} v_i)$. Observe that this is a point in the line segment between v_n and $u := \sum_{i=1}^{n-1} \frac{\lambda_i}{1-\lambda_n} v_i$. Also note that u is a convex combination of $n-1$ points, so inductively $u \in C$ which implies C contains every point in the line segment between v_n and u, and hence C contains this convex combination as desired.

9.1c *Prove that* $\mathrm{conv}(S)$ *is convex for any $S \subseteq \mathbb{R}^d$ and conclude that* $\mathrm{conv}(S)$ *is the smallest convex set containing S [1+].*

Let $x, y \in \mathrm{conv}(S)$, which by definition means we can write them as convex combinations, say with $x = \sum_{i=1}^m \lambda_i v_i$ and $y = \sum_{i=m+1}^n \lambda_i v_i$. Observe that each point on the line segment between x and y has the form $\mu x + (1-\mu)y = \sum_{i=1}^m \mu\lambda_i v_i + \sum_{i=m+1}^n (1-\mu)\lambda_i v_i$. One can check that this is a convex combination of points in S, and so we conclude that every point in the line segment between any two $x, y \in \mathrm{conv}(S)$ is in $\mathrm{conv}(S)$, proving that S is convex.

By the previous part, we have by definition of $\mathrm{conv}(S)$ that every convex set C containing S contains $\mathrm{conv}(S)$; that is, $\mathrm{conv}(S) \subseteq \bigcap_{C\in\mathcal{C}} C$ where $\mathcal{C}$ is the set of convex sets of S. By what we just proved we have $\mathrm{conv}(S) \in \mathcal{C}$, which means that $\mathrm{conv}(S) = \bigcap_{C\in\mathcal{C}} C$ and hence this is the smallest convex set containing S.

9.5a *Prove that if $S \subseteq \mathbb{R}^d$ is a set of $d+2$ points in general position, then there is a unique (unordered) partition $S_1 \cup S_2 = S$ with $\text{conv}(S_1) \cap \text{conv}(S_2) \neq \emptyset$ [2].*

If $S_1 \cup S_2 = S$ is a partition with $\text{conv}(S_1) \cap \text{conv}(S_2) \neq \emptyset$, then this implies there exists an affine dependency relationship $\sum \alpha_i x_i = 0$ with $x_i \in S_1$ if and only if $\alpha_i > 0$. Assume for contradiction there were some other partition $T_1 \cup T_2$ which has an affine dependency $\sum \beta_i x_i = 0$. Because these are distinct partitions, one can choose some ε appropriately so that $\sum \alpha_i x_i + \varepsilon \sum \beta_i x_i = 0$ is an affine dependency with at most $d+1$ non-zero coordinates, contradicting that S is in general position.